

Algebraic Topology: a beginner's course

This lecture:

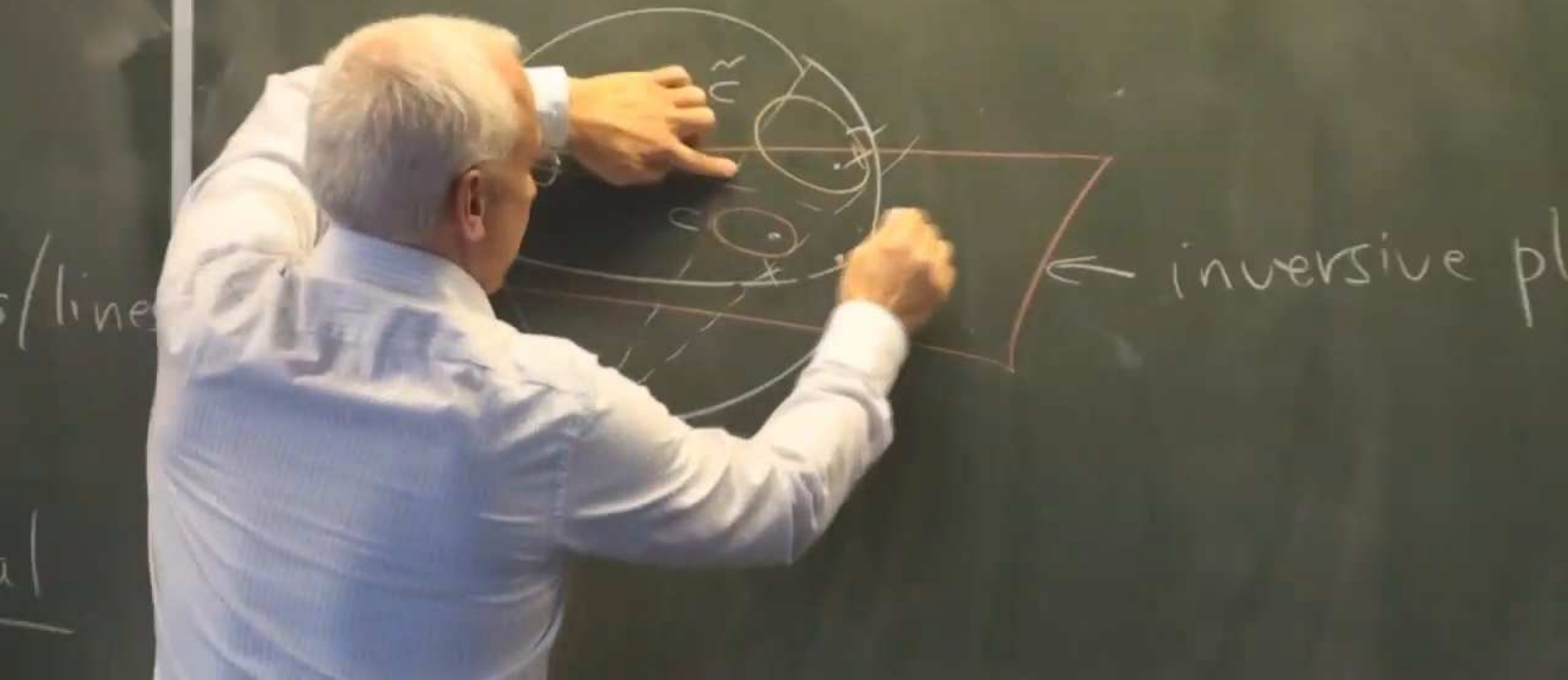
**AlgTop4:
More on the sphere**

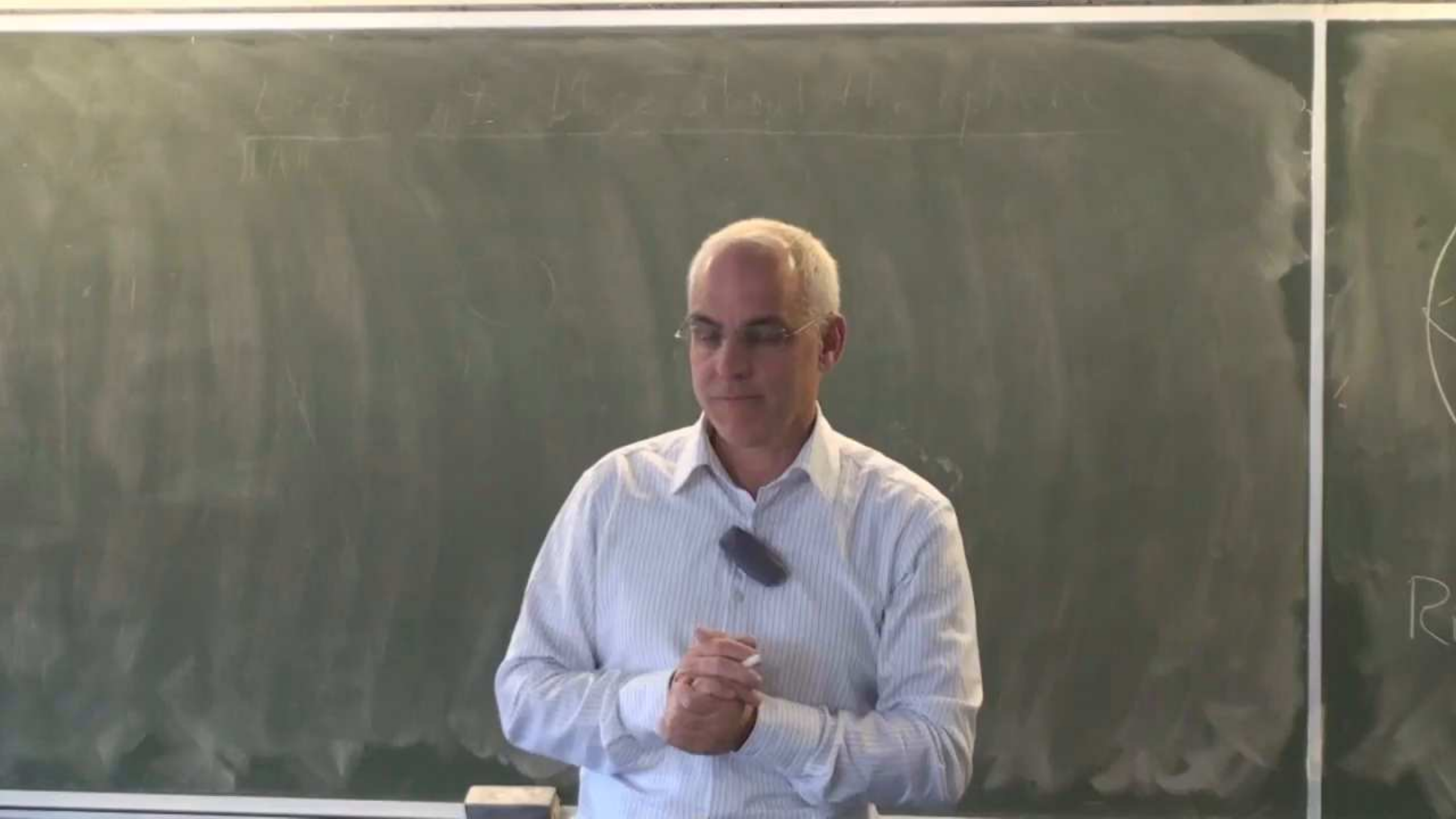
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Lecture 4: More about the sphere



Inversive plane $\cong S$

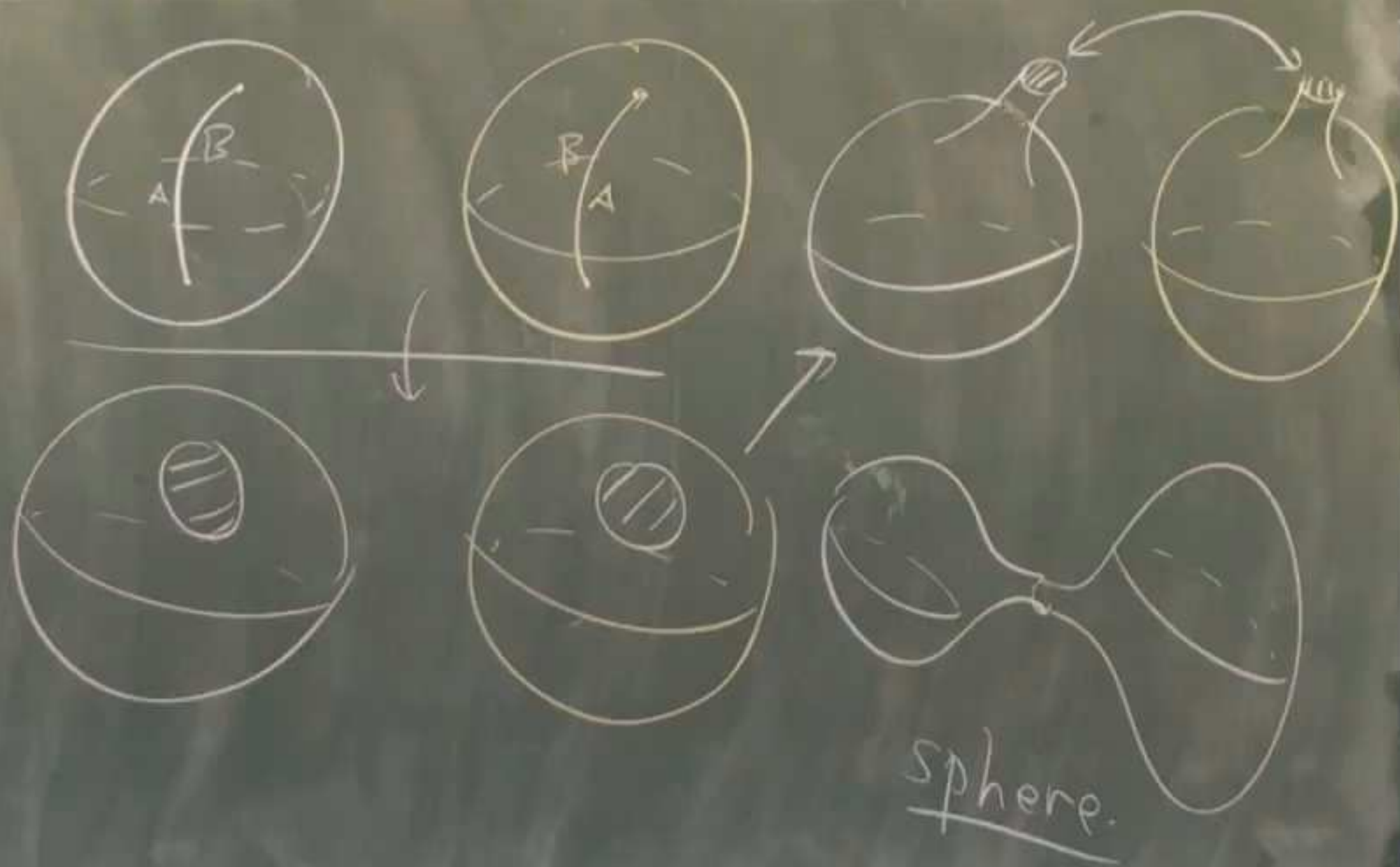




Lecture 4: The geometry of the sphere

PLAN

R



Riemann surface

$$\mathbb{C}^2 \quad [z, w] \quad z, w \in \mathbb{C}$$

$$[\vec{v}] = [\lambda \vec{v}]$$

$[z, w] = \vec{v}$ Q All 1-dim subspaces?

$$\text{If } w \neq 0 \text{ then } [(z, w)] = [(z/w, 1)]$$

$\Rightarrow$

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next in the series:

**AlgTop5: Two-dimensional
surfaces: the torus and genus**